1、解：

$\|A_1\|_1 = \max\min\{|1+2+3|, |5+0+4|, |3+2+1|\}$

$= \max\{6, 9, 6\} = 9$

$\|A_2\|_2 = \sqrt{\lambda_{max}(A^TA)}$ ⇒

$$A^TA = \begin{pmatrix} 1 & 5 & \cancel{3} \\ 2 & 0 & \cancel{2} \\ 3 & 4 & \cancel{1} \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 5 & 0 & 4 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 5 & 3 \\ 2 & 0 & 2 \\ 3 & 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 14 & 17 & 10 \\ 17 & 41 & 19 \\ 10 & 19 & 14 \end{pmatrix}$$

$$|A^TA - \lambda E| = \begin{vmatrix} 14-\lambda & 17 & 10 \\ 17 & 41-\lambda & 19 \\ 10 & 19 & 14-\lambda \end{vmatrix} = (14-\lambda)(41-\lambda)(14-\lambda) + 6460 - 100(41-\lambda) - 650(14-\lambda) = 0$$

解得 $\lambda = 6, \frac{63+3\sqrt{345}}{2}, \frac{63-3\sqrt{345}}{2}$

$\therefore \|A_2\| = \sqrt{\frac{63+3\sqrt{345}}{2}}$

$\|A\|_\infty = \max\{|1+5+3|, |2+0+2|, |3+4+1|\}$

$= \max\{9, 4, 8\} = 9$

2、解：

(1)

$\frac{\partial f(x)}{\partial x_1} = \frac{2x_1}{x_1^2+x_2+x_3^2}$ 　代入 $M(1, 2, -1)$ 得

$\frac{\partial f(x)}{\partial x_2} = \frac{2x_2}{x_1^2+x_2^2+x_3^2}$ 　$grad(f(x))|_M = (\frac{2}{9}, \frac{4}{9}, -\frac{4}{9})^T$

$\frac{\partial f(x)}{\partial x_3} = \frac{2x_3}{x_1^2+x_2^2+x_3^2}$

$\nabla f(x) = (\frac{2}{9}, \frac{4}{9}, -\frac{4}{9})^T$

(2) $H(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \frac{\partial^2 f(x)}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f(x)}{\partial x_3 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_3 \partial x_2} & \frac{\partial^2 f(x)}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} \frac{-2\cdot 2x_1}{(x_1^2+x_2^2+x_3^2)^2} & 0 & 0 \\ 0 & \frac{-2\cdot 2x_2}{(x_1^2+x_2^2+x_3^2)^2} & 0 \\ 0 & 0 & \frac{-2\cdot 2x_3}{(x_1^2+x_2^2+x_3^2)^2} \end{bmatrix}$

代入 $M(1, 2, -1)$

$$H(x) = \begin{bmatrix} -\frac{4}{81} & 0 & 0 \\ 0 & -\frac{8}{81} & 0 \\ 0 & 0 & \frac{4}{81} \end{bmatrix}.$$

3. 解: $f(x) = \log\det(X)$

$$\frac{d}{dx}\log\det(X) = \frac{1}{\det(X)} \cdot \frac{d}{dx}\det(X)$$

代入 $\frac{d}{dx}\det(X) = \det(X)\cdot(X^{-1})^T$

$$\therefore \frac{d}{dx}\log\det(x) = \frac{1}{\det(x)} \cdot \det(x)\cdot(x^{-1})^T = (X^{-1})^T$$

$$\Rightarrow \nabla_X f(X) = [(X^{-1})^T]^T = X^{-1}$$

$\therefore$ 函数$f(x) = \log\det(X)$关于矩阵X的梯度是X的逆矩阵.

4、证明:

4.1 $A, B \in S^n$，则 $A B$ 都是实对称矩阵

则 $\lambda A+(1-\lambda)B$ 也为实对称，

∴结论成立.

4.2. 命题并不成立.

反例：$A=\begin{pmatrix}1&0\\0&0\end{pmatrix}$ $B=\begin{pmatrix}0&0\\0&1\end{pmatrix}$，A.B 不满足半正定.

$\lambda=\frac{1}{2}$ 时，$\lambda A+(1-\lambda)B=\frac{1}{2}\begin{pmatrix}1&0\\0&0\end{pmatrix}+\frac{1}{2}\begin{pmatrix}0&0\\0&1\end{pmatrix}$

$=\begin{pmatrix}\frac{1}{2}&0\\0&\frac{1}{2}\end{pmatrix}$

4.3. A, B 均为正定矩阵 ⇒ 特征值都大于0.

正定矩阵的线性组合具有凸性.

特征值连续依赖于矩阵元素.

∴ $\lambda A+(1-\lambda)B$ 为正定矩阵

5. 解：$A \in S_{++}^n \Rightarrow \lambda_1, \lambda_2, \cdots \lambda_n > 0$ 且 $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_n$

$x^T A x = \sum_{i=1}^n \lambda_i (x^T e_i)^2$ e_i 为对应 λ_i 的特征向量

$\|x\|_2^2 = 1 \Rightarrow x = \sum_{i=1}^n a_i e_i$ $\sum_{i=1}^n a_i = 1$

$\therefore x^T A x = \sum_{i=1}^n \lambda_i a_i^2$

由均有不等式得，$\lambda_1 \sum_{i=1}^n a_i^2 \geq \sum_{i=1}^n \lambda_i a_i^2 \geq$ 卅 $\lambda_n \sum_{i=1}^n a_i$

代入 $\sum_{i=1}^n a$ 即 本 $\lambda_1 \geq x^T A x \geq \lambda_n$.

$\therefore x^T A x$ 的最大值为 λ_1，最小值为 λ_2

6. 解：$f_i(y) = -\log \frac{\exp(y_i)}{\sum_{j=1}^n \exp(y_j)}$ $\forall i = 1, 2, \cdots, n$

$\therefore f_i(y) = -\log \frac{1}{1+\sum_{j \neq i} \exp(y_j - y_i)} = \log[1+\sum_{j \neq i} \exp(y_j - y_i)]$

求 $\frac{\partial f_i(y)}{\partial y_i}$.

当 $i = i$ 时 $\frac{\partial f_i(y)}{\partial y_i} = \frac{-1}{1+\sum_{j \neq i} \exp(y_i - y_j)} \cdot (-\sum_{j \neq i} \exp(y_j - y_i) \cdot (-1))$

$$= \frac{-\sum_{j \neq i} \exp(y_j - y_i)}{1+\sum_{j \neq i} \exp(y_j - y_i)} = \cancel{\frac{-\sum_{j \neq i} \exp(y_j)}{\exp(y_i) + \sum_{j \neq i} \exp(y_j)}}$$

当 $i \neq i$ 时 $\frac{\partial f_i(y)}{\partial y_i} = \frac{1}{1+\sum_{j \neq i} \exp(y_j - y_i)} \cdot \exp(y_i - y_i)$

$$= \frac{\exp(y_i)}{\exp(y_i) + \sum_{j \neq i} \exp(y_j)}$$